

Exploring the Barriers to Vietnamese People's Tendency to Attend Yearly Checkups

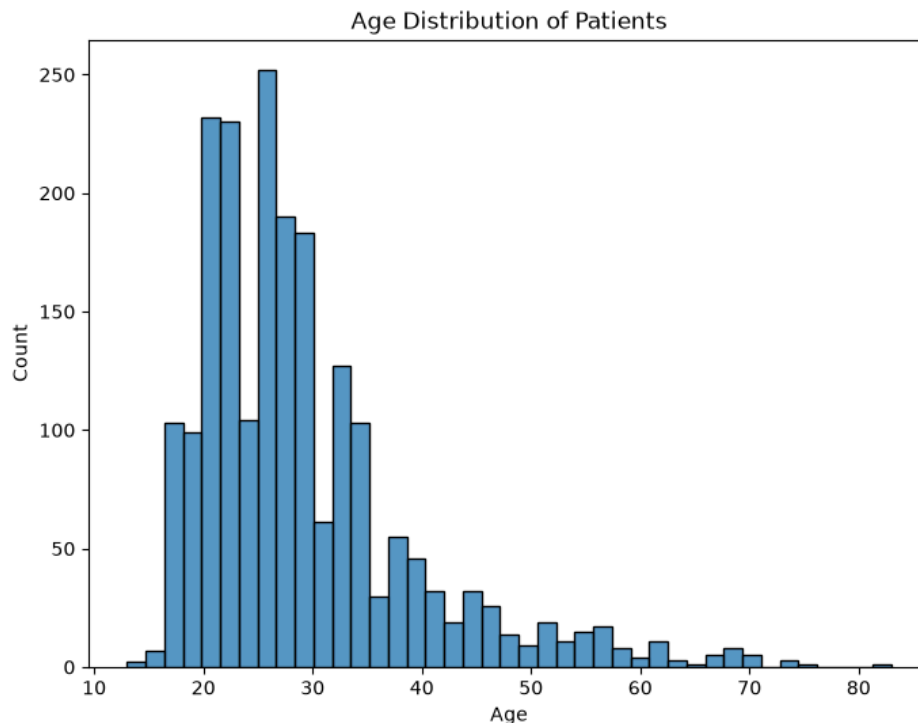
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There are many factors that shape a patient's attitudes and whether or not they make the decision to seek medical help. Third world countries like Vietnam may face different barriers that impact clinical attendance. This report explores a survey conducted in 2017 to explore Vietnamese tendency to attend annual check ups. Our goal is to explore if there are certain factors correlated to Vietnamese people's attitudes and actions in regards to getting medical help.

Who is coming to clinics?

When looking at age demographics, we see that there is a right skew. A majority of the patients are in their mid to late 20s, with a mean of about 29 years old, but a median of 27. While close in number, the average may be impacted by those in the older age. It may be better to describe the average age of participants using the median, which is more robust to outliers.

Figure 1. Patients by age group



Data Source from:

<https://cmustatistics.github.io/data-repository/medicine/vietnam-health.html#references>

Job status may be a factor that can impact annual check up attendance. Those in the average working age range mainly have stable jobs which may impact their ability to get themselves or their family members to clinics. This is because it may be hard to take off for a non-emergency procedure and the patient may not be able to afford missing a day of work. For participants in the younger (less than 18) or elderly range (older than 50), they may face troubles going to their appointments due to reliance on another family member to transport them to the clinics. These barriers may be important to consider when trying to make an impact.

Table 1. Job status by age group

Jobstt	housewife	other	retirer	stable	student	unstable
Age_gr						
<18	0.000000	0.000000	0.000000	0.000000	0.982143	0.017857
18-29	0.027200	0.037600	0.000800	0.468000	0.391200	0.075200
30-39	0.053170	0.044990	0.000000	0.820041	0.006135	0.075665
40-49	0.097403	0.103896	0.006494	0.681818	0.006494	0.103896
>=50	0.084034	0.159664	0.294118	0.268908	0.000000	0.193277
All	0.041103	0.050290	0.017892	0.543037	0.264990	0.082689

Education and health priorities

The sample is highly educated: 66.9% hold a college degree, 20.1% completed middle school, 6.9% high school, and 6.1% postgraduate study. Findings for the smaller education groups should be read with caution. Even so, the view that health is a priority is widespread and rises slightly with schooling, from 78.8% at the middle school level to 86.6% among postgraduates.

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Figure 2. Distribution of Education among participants

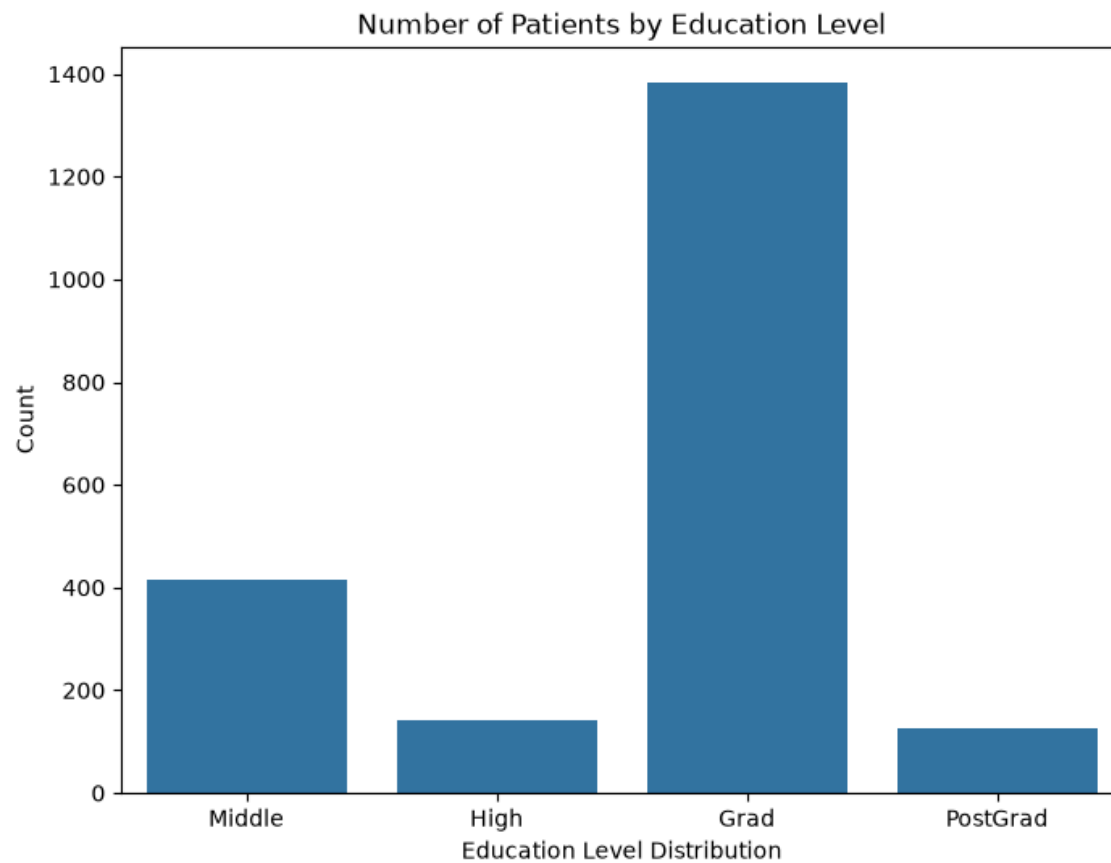


Table 2. Health seen as a priority, by education (%)

HthyPriority	no	yes
Edu		
Middle	21.153846	78.846154
High	19.718310	80.281690
Grad	18.799711	81.200289
PostGrad	13.385827	86.614173

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Differing Attitudes by Location

A majority of the respondents come from the capital of Vietnam, Hanoi. This is about 79% of the survey (1,638 participants). Only 13 locations have ten or more responses, so comparisons are limited to those. Thanh Hoa has the highest share saying health is not a first priority (44%), however it is important to note that this location only contains a sample size of 18 respondents. Checkups are widely seen as a waste of time, with seven of the thirteen locations above 60%.

Table 3. Attitudes by location (places with 10+ responses)

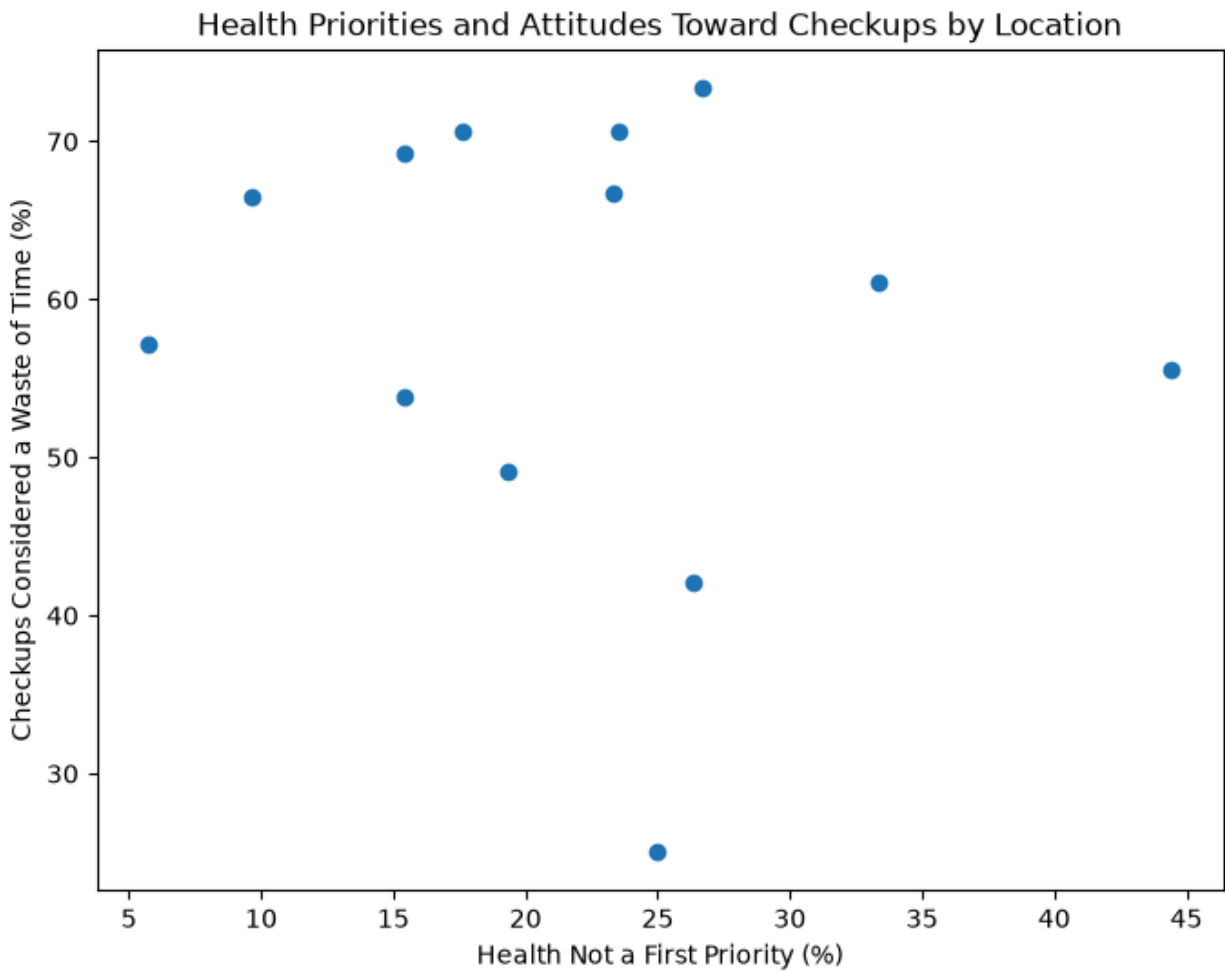
Place	Responses	Not first priority (%)	Checkup a waste of time (%)
Hanoi	1,638	19.4	49.1
Hung Yen	155	9.7	66.5
Ho Chi Minh	35	5.7	57.1
Nam Dinh	30	23.3	66.7
Phu Tho	19	26.3	42.1
Bac Ninh	18	33.3	61.1
Thanh Hoa	18	44.4	55.6
Hai Duong	17	23.5	70.6
Ha Nam	17	17.6	70.6
Bac Giang	15	26.7	73.3
Ninh Binh	13	15.4	69.2
Quang Ninh	13	15.4	53.8
Thai Binh	12	25.0	25.0

Across these locations, the correlation between “not a priority” and “waste of time” is weak and negative ($r = -0.15$, $n = 13$), so the two attitudes do not move together. People may value health yet still doubt the value of routine checkups. That gap is something outreach can address.

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Figure 3. Correlation between health priorities and attitudes towards checkups ($r = -0.15$)



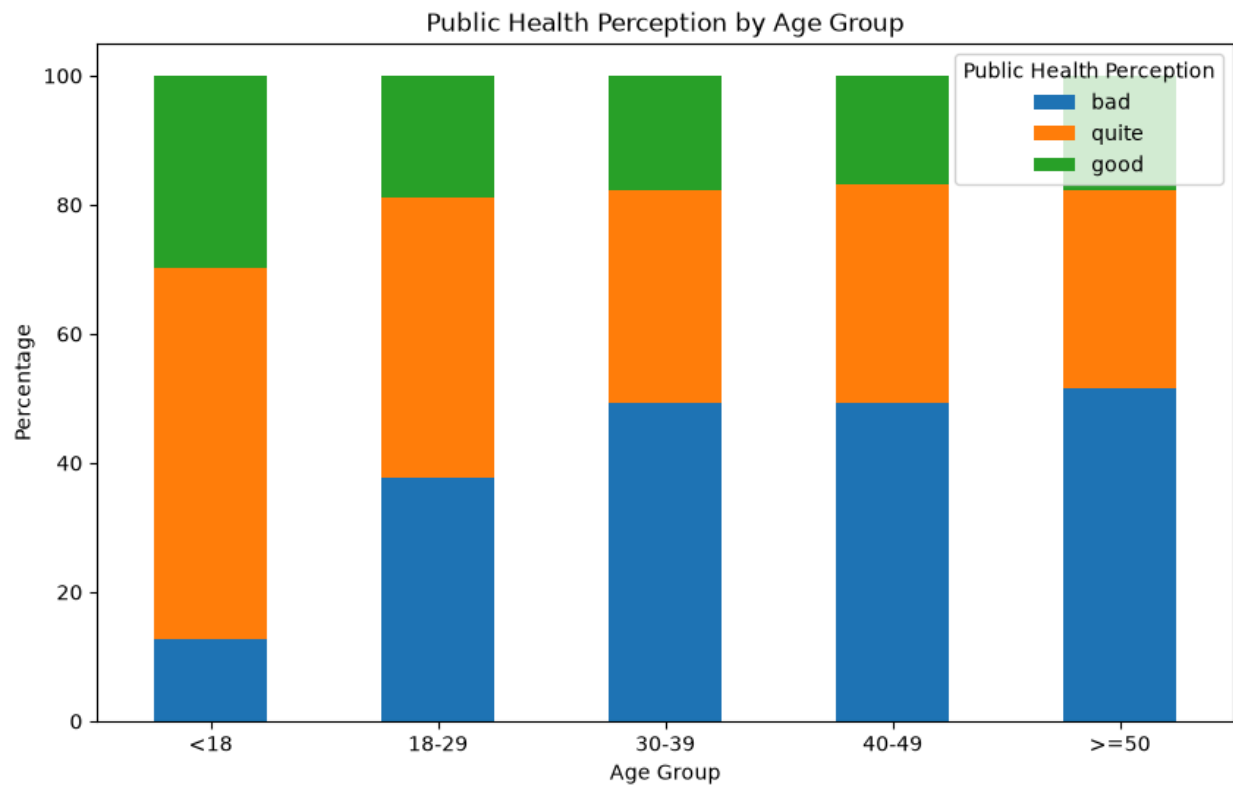
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Age and trust in public health

Perception of public health worsens with age. Only 12.8% of under-18s rate it “bad”, compared with 37.7% of 18–29s and about half of everyone aged 30 and over. It may be important to target public health education or changes towards the needs of those in older age groups.

Figure 3. Public health perception by age group (%)



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Could IT help?

Willingness to use IT is mixed. In every group but one, “yes” is the most common answer. The exception is high school graduates, where 44.4% say no and only 22.5% say yes. Interest in acting on an app's checkup reminder is more cautious: “maybe” is the top answer in every group.

Figure 4. Willingness to use IT by education level (%)

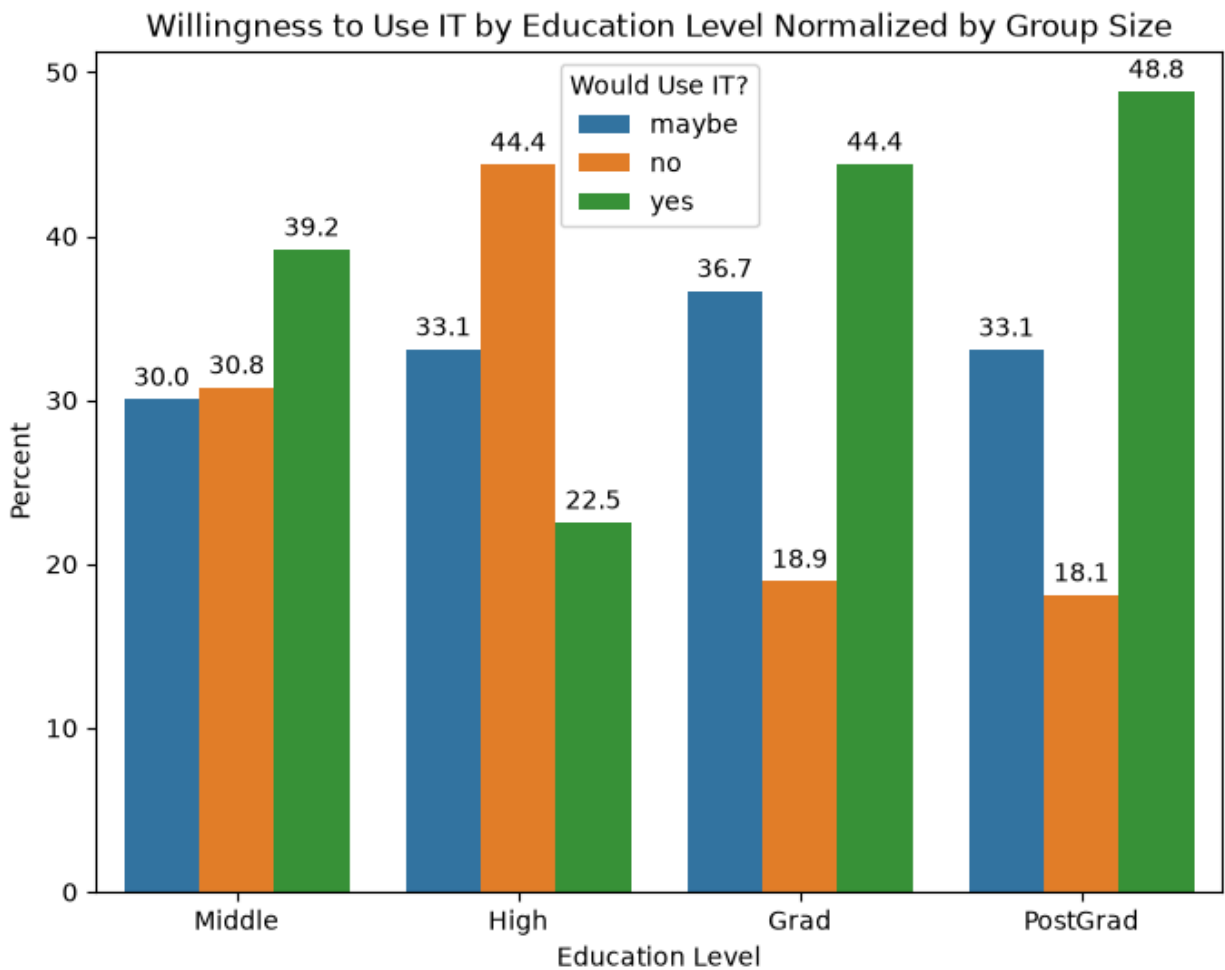


Table 4. Response to an app-prompted checkup, by education (%)

Education	Maybe	No	Yes
Middle	41.3	19.5	39.2
High	39.4	37.3	23.2
Grad	44.5	14.8	40.7
PostGrad	44.9	11.0	44.1

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Recommendations

While our analyses were limited and unable to make significant conclusions towards factors that contribute to Vietnamese people's perceptions of health and willingness to attend yearly checkups, some of our findings may reveal some possible changes that could be made to better support the Vietnamese people. Since many of the older generation have negative perceptions toward public health, it may be important for the government to rebuild their trust. Whether it be through creating social programs to address their needs or providing more age appropriate support. Next, it may be important to target some age groups that may need more support getting to the clinics. For example, children and elderly people who may need guidance getting to the clinics. Maybe getting at home care visits or implementing digital checkups. In this scope, it may be helpful to implement some IT related solutions to make healthcare more accessible for their yearly checkups. Since some participants were hesitant to use these tools, education towards these technologies may make them more willing to try out these solutions. Ultimately, it is important to note that our sample size is limited to different cities and age groups. Some demographics may be more underrepresented or overrepresented than others, making it difficult to generalize to everyone.

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